Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/13/2013
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Investigations of the Plant Cytoskeleton in Microgravity With Gene Profiling and Cytochemistry		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences BRIC NNH09ZTT004N
Start Date:	02/08/2010	**End Date:**	02/07/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-131 NOTE: extended to 2/7/2013 per NSSC information (Ed., 1/27/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX10AF44G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goal of this research is to study the effects of microgravity on the structure and organization of actin cytoskeleton in plants. This proposed project builds on our previous ground-based and spaceflight research using the model plant Arabidopsis. Thus, the specific aims of this proposed flight research using the BRIC-LED system are: (1) to investigate plastid position in statocytes (gravity-perceiving cells) in microgravity; (2) to determine the effect of microgravity on the actin cytoskeletal organization in gravity-perceiving cells; (3) to study microgravity effects on actin cytoskeleton-related gene expression in plant cells. This project also will allow us to directly correlate results from cytological investigations and gene profiling in order to understand the nature of the actin cytoskeleton in mechanisms of gravity perception. This proposed research focuses on the effects of gravity on basic cellular mechanisms and processes in plants, so it is relevant to the emphasis on using Arabidopsis as a model system in developmental biology highlighted in the current NRA. The PI's group has extensive experience with the use of Arabidopsis in experiments on the Space Shuttle and the ISS. Preliminary data on the growth of seedlings in the BRIC-LED system and from the gene profiling studies performed during our recent experiments on the ISS are presented to show the feasibility of the proposed project. Improved knowledge of the basic mechanistic processes that will be the focus of this research is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposed project on the analysis of the role of the actin cytoskeleton in mechanisms of gravity perception builds on our previous ground- and flight-based research. We plan to study the effects of gravity on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology as highlighted in the current NASA Research Announcement (NRA). Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our spaceflight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture.
Task Progress:	In plants, sensitive and selective mechanisms have evolved to perceive and respond to light and gravity. We investigated the effects of microgravity on the growth and development of Arabidopsis thaliana (ecotype Landsberg) in a spaceflight experiment. These studies were performed using the Biological Research in Canisters (BRIC) hardware system in the middeck region of the Space Shuttle during mission STS-131 in April 2010. Seedlings were grown on nutrient agar in Petri dishes in BRIC hardware under dark conditions and then were fixed in-flight with paraformaldehyde, glutaraldehyde, or RNAlater®. Although the long-term objective was to study the role of the actin cytoskeleton in gravity perception, in this paper, we focus on the analysis of morphology of seedlings that developed in microgravity. While previous spaceflight studies noted deleterious morphological effects due to the accumulation of ethylene gas, no such effects were observed in seedlings grown using the BRIC system. Seed germination was 89% in the spaceflight experiment and 91% in the ground control, and seedlings grew equally well in both conditions. However, roots of space-grown seedlings exhibited a significant difference (compared to the ground controls) in overall growth patterns in that they skewed to one direction. In addition, a greater number of adventitious roots formed from the axis of the hypocotyls in the flight-grown plants. Our hypothesis is that an endogenous response in plants causes the roots to skew and that this default growth response is largely masked by the normal 1-g conditions on Earth. In the last year, we continued morphometric studies of the space-grown seedlings with light and electron microscopy. We also focused on gene profiling studies by using microarray techniques.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Johnson CM, Edelmann RE, Kiss JZ. "Analyses of statoliths in petioles of seedlings grown in microgravity." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 101. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Johnson CM, Millar KD, Edelmann RE, Kiss JZ. "Morphometric studies of the endodermis in petioles of seedlings grown during a spaceflight experiment." Microscopy & Microanalysis 2013 Meeting, Indianapolis, IN, August 4-8, 2013. Abstract Book. Microscopy & Microanalysis 2013 Meeting, Indianapolis, IN, August 4-8, 2013. In press as of March 2013. , Mar-2013
Articles in Peer-reviewed Journals	Kiss JZ. "Plant biology in reduced gravity on the Moon and Mars." Plant Biol (Stuttg). 2014 Jan;16 Suppl 1:12-7. Review. Epub 2013 Jul 25. https://dx.doi.org/10.1111/plb.12031 ; PubMed PMID: 23889757 [Note reported originally in March 2013 as "in press"] , Jan-2014
Articles in Peer-reviewed Journals	Johnson CM, Subramanian A, Pattathil S, Correll MJ, Kiss JZ. "Comparative transcriptomics indicate changes in cell wall organization and stress response in seedlings during spaceflight." American Journal of Botany. 2017 Aug;104(8):1219-31. https://doi.org/10.3732/ajb.1700079 ; PubMed PMID: 28827451; PubMed Central PMCID: PMC5821596 , Aug-2017
Articles in Peer-reviewed Journals	Kiss JZ, Aanes G Schiefloe M, Coelho LH, Millar KD, Edelmann RE. "Changes in operational procedures to improve spaceflight experiments in plant biology in the European Modular Cultivation System." Advances in Space Research. 2014 Mar;53(5):818–27. http://dx.doi.org/10.1016/j.asr.2013.12.024 , Mar-2014
Awards	Kiss JZ. "Benjamin Harrison Medallion (named for the 1852 Miami graduate who was U.S. President). This highest award to a faculty member recognizes outstanding national contributions to education and other international recognition beyond his field, May 2012." May-2012